

20260929 Pawsey Computing and Storage

This estimate uses **one 4800-8900 Angstrom nGIST pass** over 39 datacubes representing 40 MAUVE-MUSE galaxies. The full pass is estimated at **1,143.018 node-hours**, or **146,306 SU** on 128 physical cores. A single 20% buffer allowance brings the reference to **175,568 SU per pass**. The number of future science passes remains open.

Measured 8900-Angstrom runs

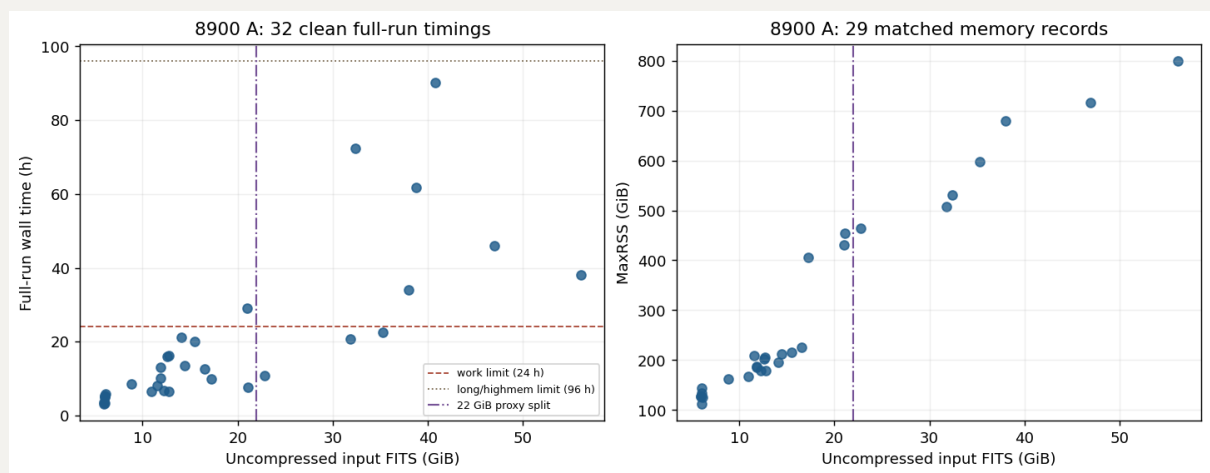


Figure 1. The 32 blue timing points are clean, complete 8900-Angstrom runs; 29 have matched peak-memory records. The 22 GiB line is a rough input-size split for estimating missing runtimes, not the Setonix queue-assignment rule.

The measured small group (<22 GiB; 23 cubes), which is able to run under **work** or **long** queue on Pawsey, has a maximum of **29.170 hours** (NGC4396); the large group (>=22 GiB; 9 cubes), which requires to run under **highmem** queue on Pawsey, has a maximum of **90.253 hours** (NGC4654).

Both maxima come from **8900-Angstrom runs**. An untimed cube receives the largest observed runtime in its own input-size group.

Complete 39-cube reference pass

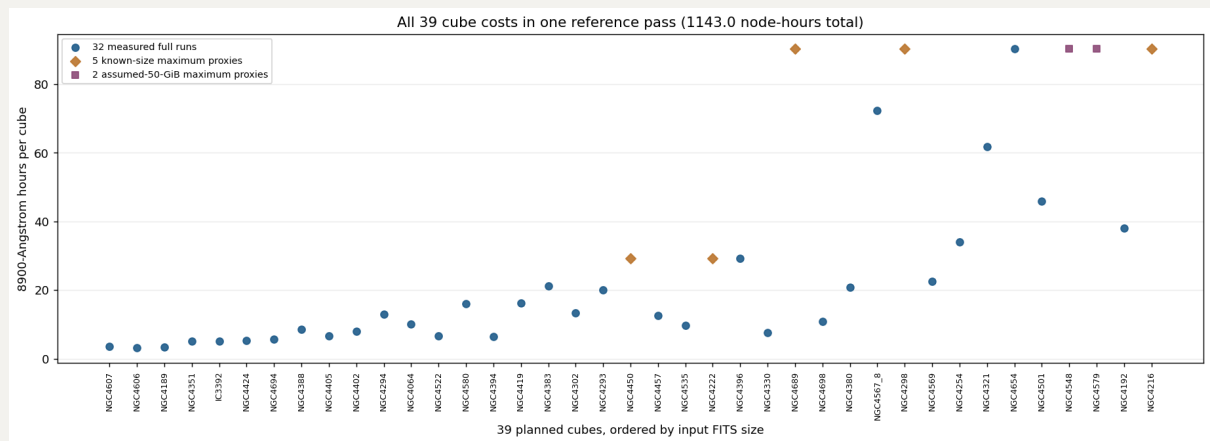


Figure 2. All 39 cubes enter the budget: 32 measured timings (blue), five known-size maximum-based estimates (orange), and two assumed-size maximum-based estimates (purple). Check out Appendix A. for more details.

Note that the two missing datacubes so far, NGC4548 and NGC4579, are each assumed to be **50 GiB**. The estimates are labelled separately from measurements.

CONTRIBUTION	CUBES	NODE-HOURS
Measured complete runs	32	633.411
Known-size small/large maximum estimates	5	329.101
Two assumed-50-GiB large-cube maximum estimates	2	180.507
One complete pass	39	1,143.018

Convert to SU and scale later

Pawsey counts one requested physical CPU core-hour as one SU. Each job requests one **128-physical-core** node, so one node-hour costs **128 SU**. Using the unrounded cube timings:

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1,143.018 node-hours x 128 physical cores = 146,306 SU
20% allowance                               = 29,261 SU
Buffered reference for one pass              = 175,568 SU
For N comparable passes: SU(N)              = N x 175,567.53

```

So far for 35 completed datacubes they occupy 8TB space and so we expect **8 x 39/35 = 8.914 TB** of output for one 39-cube pass, plus **1.5 TB** for all input cubes and other necessary ingredients. If **N** output families must be retained together, **Acacia minimum(N) = 1.5 + N x (8 x 39/35) TB**. One family needs **10.414 TB** before headroom, or **15 TB** after 15% headroom and upward rounding.

The 20% allowance and the two missing cube sizes are purely based on my personal assumptions. Each group maximum is conservative within the measured sample, but it is not a guaranteed upper bound for an unmeasured cube.

Appendix A. Per-cube 8900-Angstrom reference list

This table is generated from the same unrounded calculations. `Full LOGFILE` is a clean complete attempt; `Small-cube maximum` and `Large-cube maximum` use the largest measured runtime in the matching 8900-Angstrom size group. Input size is uncompressed FITS **GiB**; the two assumed sizes are **50 GiB each**.

CUBE ID	INPUT SIZE (GiB)	SIZE GROUP	RUNTIME BASIS	HOURS	SU AT 128 CORES
NGC4607	5.94	small (<22 GiB)	Full LOGFILE	3.54	453
NGC4606	5.95	small (<22 GiB)	Full LOGFILE	3.22	413
NGC4189	6.01	small (<22 GiB)	Full LOGFILE	3.46	443
NGC4351	6.02	small (<22 GiB)	Full LOGFILE	5.03	644
IC3392	6.03	small (<22 GiB)	Full LOGFILE	5.11	654
NGC4424	6.04	small (<22 GiB)	Full LOGFILE	5.40	691
NGC4694	6.10	small (<22 GiB)	Full LOGFILE	5.75	736
NGC4388	8.86	small (<22 GiB)	Full LOGFILE	8.51	1,089
NGC4405	10.91	small (<22 GiB)	Full LOGFILE	6.56	840
NGC4402	11.55	small (<22 GiB)	Full LOGFILE	7.99	1,022
NGC4294	11.85	small (<22 GiB)	Full LOGFILE	12.97	1,660
NGC4064	11.87	small (<22 GiB)	Full LOGFILE	10.15	1,300
NGC4522	12.26	small (<22 GiB)	Full LOGFILE	6.72	860

		GiB)			
NGC4580	12.63	small (<22 GiB)	Full LOGFILE	16.07	2,057
NGC4394	12.73	small (<22 GiB)	Full LOGFILE	6.51	833
NGC4419	12.75	small (<22 GiB)	Full LOGFILE	16.30	2,086
NGC4383	14.07	small (<22 GiB)	Full LOGFILE	21.25	2,719
NGC4302	14.41	small (<22 GiB)	Full LOGFILE	13.39	1,714
NGC4293	15.48	small (<22 GiB)	Full LOGFILE	20.03	2,564
NGC4450	16.45	small (<22 GiB)	Small-cube maximum	29.17	3,734
NGC4457	16.57	small (<22 GiB)	Full LOGFILE	12.61	1,614
NGC4535	17.27	small (<22 GiB)	Full LOGFILE	9.80	1,255
NGC4222	20.99	small (<22 GiB)	Small-cube maximum	29.17	3,734
NGC4396	21.02	small (<22 GiB)	Full LOGFILE	29.17	3,734
NGC4330	21.11	small (<22 GiB)	Full LOGFILE	7.53	964
NGC4689	22.54	large (>=22 GiB)	Large-cube maximum	90.25	11,552
NGC4698	22.79	large (>=22 GiB)	Full LOGFILE	10.90	1,396
NGC4380	31.83	large (>=22 GiB)	Full LOGFILE	20.78	2,659
NGC4567_8	32.42	large (>=22 GiB)	Full LOGFILE	72.30	9,254
NGC4298	32.68	large (>=22 GiB)	Large-cube maximum	90.25	11,552
NGC4569	35.27	large (>=22 GiB)	Full LOGFILE	22.46	2,875

NGC4254	37.96	large (≥ 22 GiB)	Full LOGFILE	34.03	4,355
NGC4321	38.77	large (≥ 22 GiB)	Full LOGFILE	61.72	7,900
NGC4654	40.81	large (≥ 22 GiB)	Full LOGFILE	90.25	11,552
NGC4501	46.95	large (≥ 22 GiB)	Full LOGFILE	45.91	5,877
NGC4548	50.00	large (≥ 22 GiB)	Assumed 50 GiB; large maximum	90.25	11,552
NGC4579	50.00	large (≥ 22 GiB)	Assumed 50 GiB; large maximum	90.25	11,552
NGC4192	56.08	large (≥ 22 GiB)	Full LOGFILE	37.98	4,862
NGC4216	58.55	large (≥ 22 GiB)	Large-cube maximum	90.25	11,552